

their respective keys and powerful computing systems to run algorithms that solve very complex mathematical problems. When a problem is solved, the block is added to the chain and the data it contains exists on the network for ever. At this stage, the data can't be altered or removed.

Blockchain technology provides one of the best tools we currently have to protect data from hackers, preventing potential fraud and decreasing the chance of data being stolen or put into danger.

In order to destroy or corrupt a blockchain, a hacker would have to destroy the data stored on every user's computer in the global network. This could include millions of computers, with each one storing a copy of some part of or all the data. Unless the hacker could simultaneously bring down an entire network (which is nearly impossible), the undamaged computers, also known as nodes, will continue running to verify and keep a record of all the data on the network. The impossibility of a task, like taking down a whole chain, increases along with the number of users on a network. Bigger blockchain networks with more users have a lower risk of being hacked.

How does blockchain provide cyber security?

This complex structure of blockchain technology makes it the most secure. That's why innovators have begun applying blockchain technology in different sectors to prevent fraud and increase the security of data.

Bitcoin is booming, but the blockchain is evolving. Bitcoin is always a point of reference for blockchain technology. But blockchain is much more than just Bitcoin.

In terms of security, blockchain and its public ledger have potential solutions for Big Data processing, end point protection software, financial tools and supply chain management software.

Blockchain technology has impacted the cyber security industry in a few ways. The HYPR Corp is a New York based company that provides enterprises with decentralised authentication solutions, which enable consumers and employees to securely and seamlessly access mobile, Web and Internet of Things (IoT) applications. It uses blockchain technology to decentralise credentials and biometric data to facilitate risk based authentication. It invested US\$ 10 million in 2018 on this platform. NuCypher is another blockchain security company which uses distributed blockchain systems proxy re-encryption. It also has an accessible control platform and uses public-key encryption to securely transfer data and enforce access requirements.

Blockchain is one of the biggest tech buzzwords in the last few years, and the technology is being marketed as a cure for everything including cyber security. The US Ministry of Internal Affairs and Communications implemented a blockchain based system for processing government tenders in March 2018.

Every user on the blockchain has a private and public key. Any user who wants to transact, creates a digital signature using the private key along with the data to be sent. This newly created digital signature is sent along with the transaction to the receiver. The receiver has the sender's public key, using which the user can be verified.

Blockchain is like the Swiss knife for all our cyber security concerns. Financial transactions are typically guaranteed by a trusted third party (such as PayPal) and the blockchain can be used to automate that process, reducing overall costs by cutting out the middle man with autonomous smart contracts and by acting as a trusted intermediary between parties on the network.

Amid the growing adoption of blockchain, the US based Defense Advanced Research Projects Agency (DARPA) is also funding and backing the use of blockchains to help secure highly sensitive data pertaining to everything from nuclear weapons to military satellites. Stocks, mutual funds, bonds and pensions may be one day stored in a blockchain, as many financial organisations are exploring this new technology.

The cyber security industry today is significantly impacted by blockchain technology. By removing much of the human element from data storage, the blockchain significantly mitigates the risk of human errors, which is the largest cause of data breaches. It is a decentralised distributed ledger system which is used to record data transactions across multiple computers. Blockchain technology can be used to prevent any type of data breaches, identity theft cyber attacks or foul play in transactions. This ensures that the data remains private and secure.

A few ways in which blockchain can aid in cyber security are given below.

- Protected edge computing with authentication: Blockchain can help in strengthening authentication, can enhance data attribution and upgrade record management.
- Advanced confidentiality and data integrity.
- Secure private messaging.
- Improved public key infrastructure.
- Intact Domain Name System (DNS): The DNS is an easy target for malicious activities as hackers can bring down the DNS service providers for large websites like Twitter, PayPal and other services. A blockchain approach to strong DNS entries can improve the security extensively because it removes that single target which can be compromised.
- Diminished DDoS attacks: A Distributed Denial of Service (DDoS) attack is one in which multiple compromised computer systems attack a target, such as a server, website or another network resource and cause a denial of service for users' targeted resources. This forces the system to slow down or even crash, thereby denying service to legitimate users or systems. This problem can be solved by integrating the blockchain into decentralised solutions, which can provide protection against such attacks.